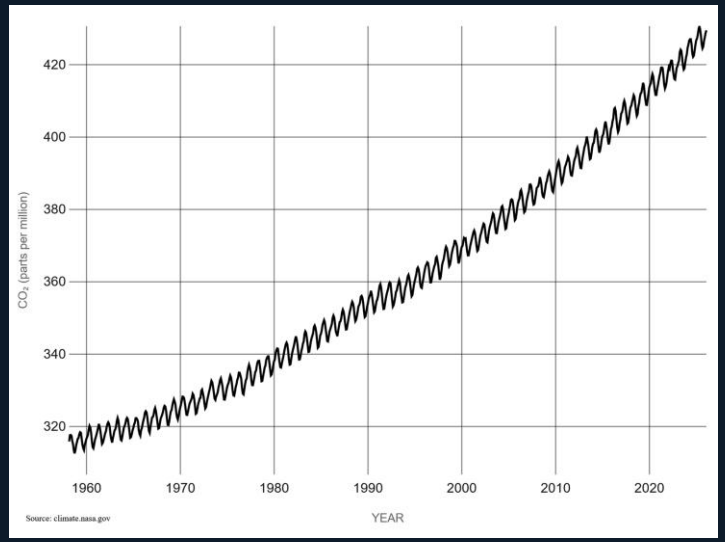
Workplace Air Quality

Monitoring

An IoT Solution using ESP32 & LoRa



CO2 concentration in the atmosphere over the years, in parts per million.

https://science.nasa.gov/earth/explore/earth-indicators/carbon-dioxide/

LoRa

SCD41

PROBLEM

CO_2 Buildup

- A CO2 level above 1,000 ppm inside buildings is an indicator of bad air quality[1] and can cause mild discomfort.
- 5000ppm is the maximum threshold for a 8hr exposure[2].
- There is no agreement on exact thresholds, different guidelines are present[3].

Temperature

Deviations from the 20–24°C optimal range can reduce productivity and physical comfort for prolonged occupancy.

Humidity

The 40–60% RH range is considered optimal for human comfort.

1. Spanish government guidelines for building ventilation
2. UK's HSE guidelines for workplaces
3. ISPRA presentation on radon and indoor air quality

APPROACH

Why This Demands an IoT Solution

Continuous Sensing

Automated 24/7 monitoring across multiple rooms simultaneously. Human intervention not needed (except for programmed battery recharging).

Real-Time Alerting

Crossing thresholds repeatedly alerts the central building management system.

Remote Aggregation

LoRa uplink sends readings to a central server for logging, dashboards, and building-wide analytics.

Edge Intelligence

The device can issues pre-alerts before an unsafe CO^2 threshold is reached, even if there is no connection to a central server.

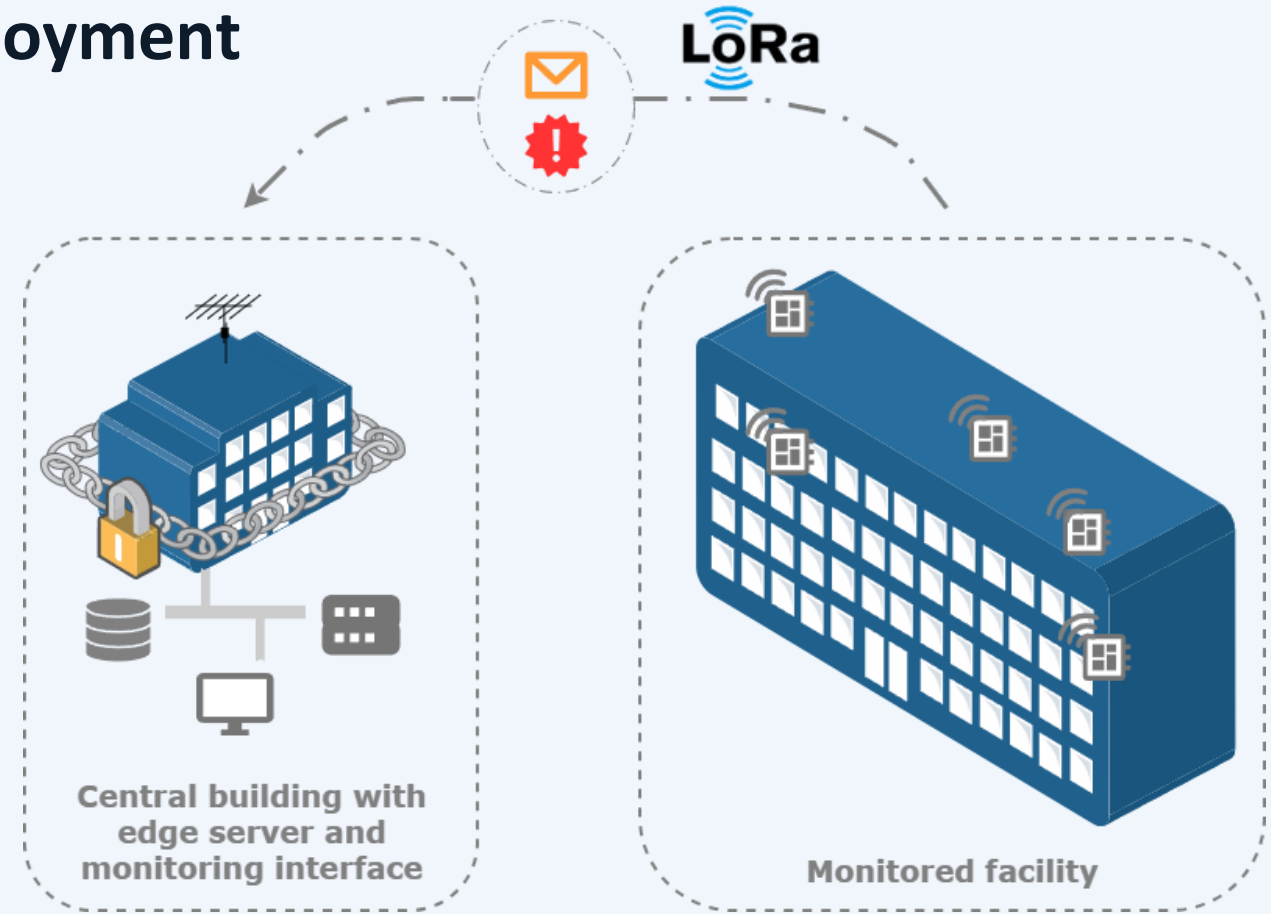
Ease of installation

The devices are fully battery operated and LoRa allows for reliable connections over long distances.

Affordability

Low component cost makes deployment practical even for large facilities.

Example deployment



SENSOR

Sensirion SCD41

CO_2 Range	400 – 5,000 ppm
Temperature	−10°C to +60°C (±0.8°C)
Humidity	0–100% RH (±6% RH)
Interface	I^2C
Power Consumption	Adjustable via software

Data Transmitted

CO_2	Parts per million (ppm)
Temperature	°C — comfort indicator
Humidity	% RH — comfort and air quality indicator
Timestamp	
Node ID	Device identifier for multi-room deployments.
Battery	Estimated battery voltage

Sensirion's SCD41 datasheet

INTELLIGENCE

Data Processing

On-Device (ESP32S3)

- **Data visualization**
  Provide simple analytics on air quality on the device's screen (on user's demand).
- **Simple status updates**
  Air quality values mapped to easily understandable color and sound cues.
- **Pre-alert logic**
  If projected trend crosses safety threshold within a certain time → issue pre-alert.
- **Adaptive sampling**
  Sampling frequency is higher when CO_2 levels are above the optimal range, otherwise sampling frequency is greatly reduced to save power.

Server-Side Analytics

- **Time-series storage**
  Historical plots of CO_2, temp, humidity per room and per node.
- **Occupancy detection**
  Link CO_2 levels to room occupancy. Automatically activate air ventilation or AC where necessary.
- **Multi-node dashboard**
  Building-wide overview with per-node status.

ACTUATORS

LED Array and sound cues

GREEN LED < 800 ppm

Air quality is good — no action needed.

AMBER LED 800–1,200 ppm

Ventilation recommended.

RED LED >>> 1,200 ppm for a prolonged amount of time

Ventilation strongly recommended.

Buzzer — Audible Alarm

Snooze

Button press silences alarm for a configurable period to prevent alarm fatigue. The use of this feature is notified to the central server.

Pre-Alert Tone

Intermittent pattern played before crossing the danger threshold. Distinguishable from full alarm.

Full Alarm

Continuous tone when CO_2 exceeds danger threshold (close to 5000ppm). Different from the pre-alert tone and should induce urgency. An alert message is sent to the central server.

CONSTRAINTS

Latency

The monitored issue doesn't require low latency.

- The measured phenomenon is slow (RH, temperature and CO_2 variations).
- If a danger threshold is reached, the device itself can react by alerting the users.

Bandwidth

- LoRa throughput is low (0.3 – 50 kbps) but sufficient for the task.
- The indoor sensors cannot be too far (<2km) from the central building with the LoRa receiver, unless a LoRa signal repeater is installed.

Sampling Frequency

- CO2, RH and temperature dynamics are slow.
- The SCD41 supports 4 operation modes:

high performance (5 sec sampling period), low power (30 sec sampling period), idle single shot, power cycled single shot.

- We aim for low power, or idle single shot (once every 5 mins) and change to adaptive mode (every min) if threshold is crossed.

Power Consumption

Most important constraint, given that there is no way to harvest energy from the environment.

- Change battery every 3 months.

Error Rate: Aim is to reduce FPR such that is it close to 0, to prevent habituation to false alarms.

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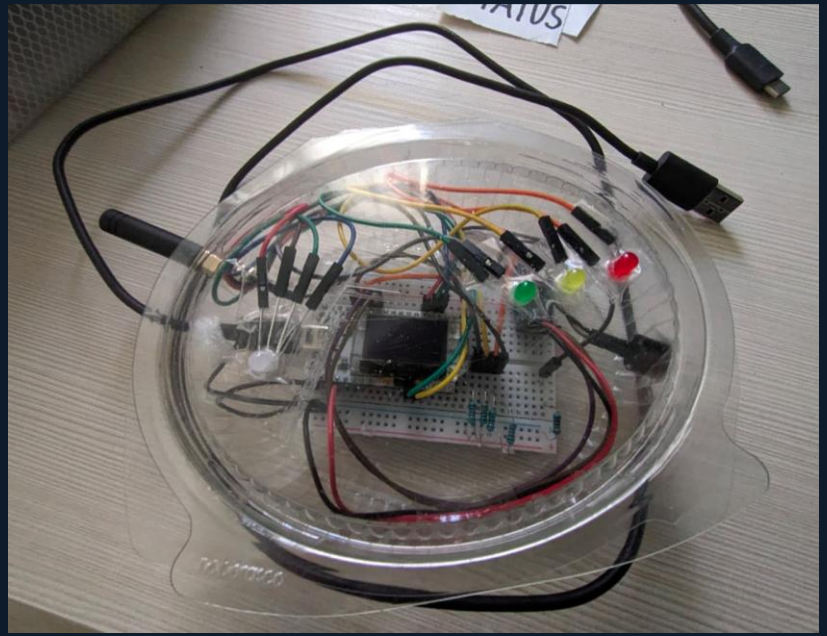
2nd Delivery – 10/04/2026



PROBLEM RECAP

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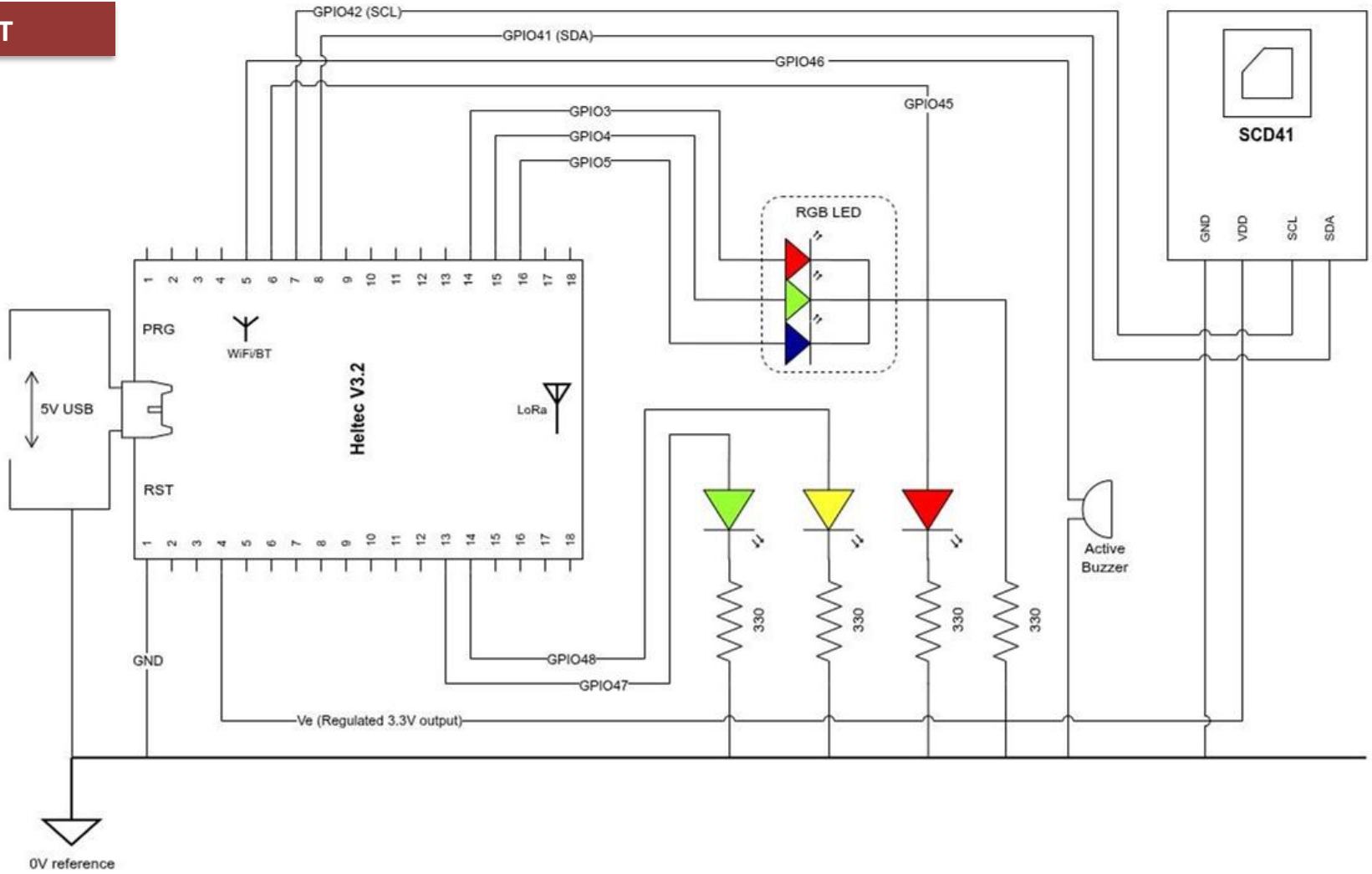
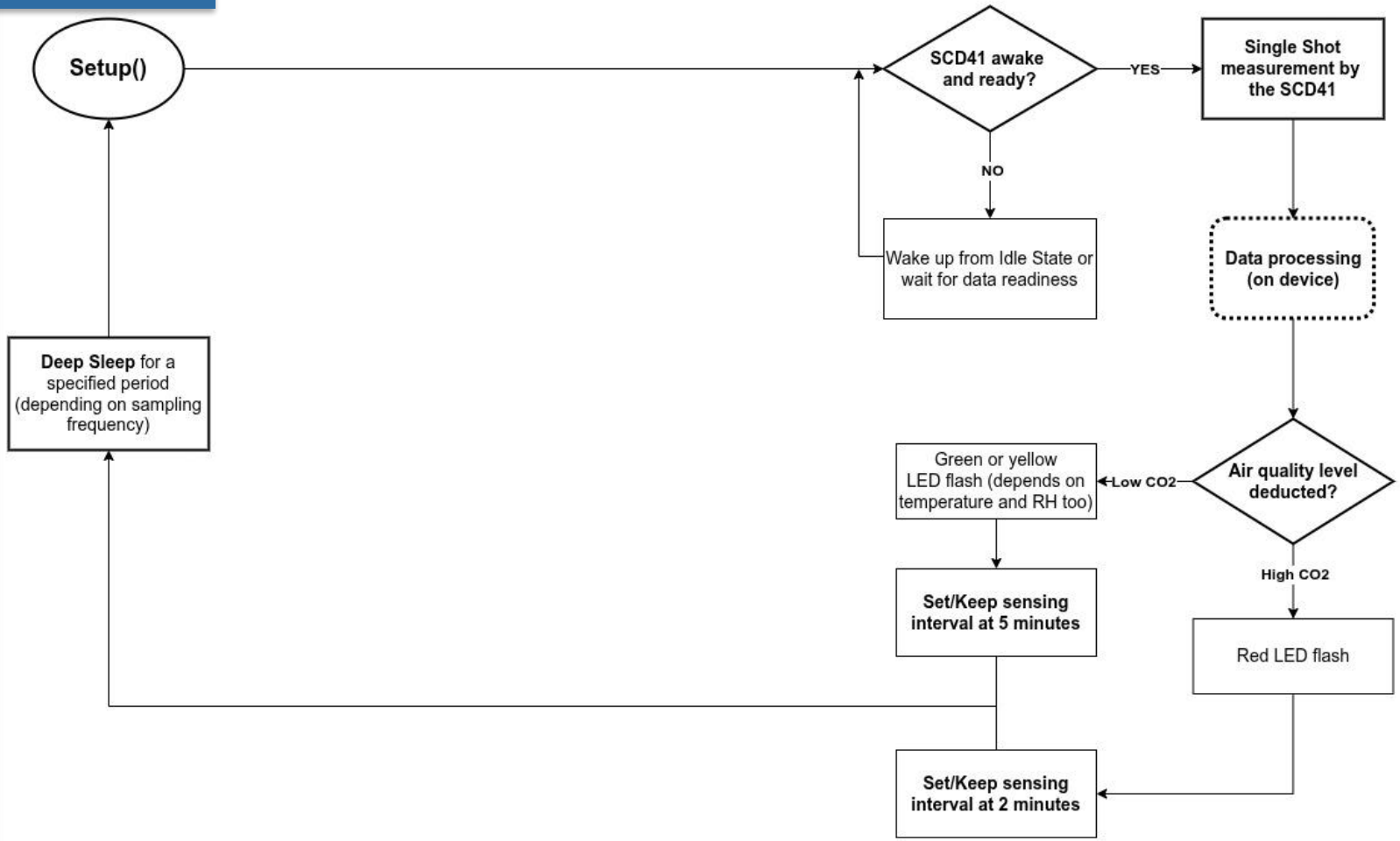
CIRCUIT


GPIO42 (SCL)
GPIO41 (SDA)
GPIO46
GPIO45
GPIO3
GPIO4
GPIO5
SCD41
GND
VDD
SCL
SDA
RGB LED
PRG
WiFi/BT
Heltec V3.2
LoRa
5V USB
RST
Active Buzzer
330
330
330
330
GND
GPIO48
GPIO47
Ve (Regulated 3.3V output)
0V reference

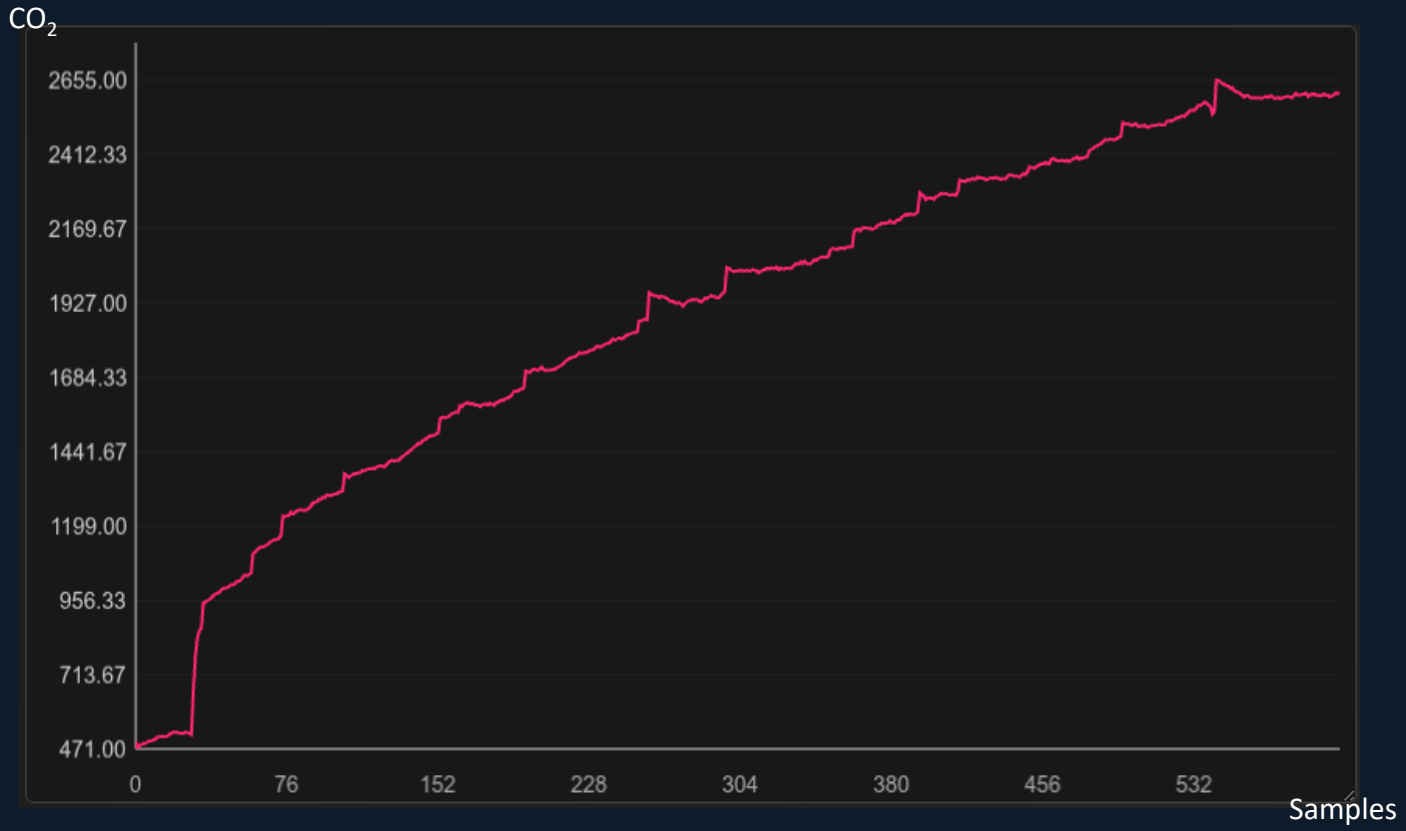
HIGH LEVEL PROGRAM



VIDEO

EXPERIMENT

- The sensor is exposed to fresh air from outside.
- Then the sensor is deployed inside a room, with two people; the room's door and windows are kept closed for the experiment duration.
- The experiment duration was about 5 hours.
- The interval between samples is 30 seconds.



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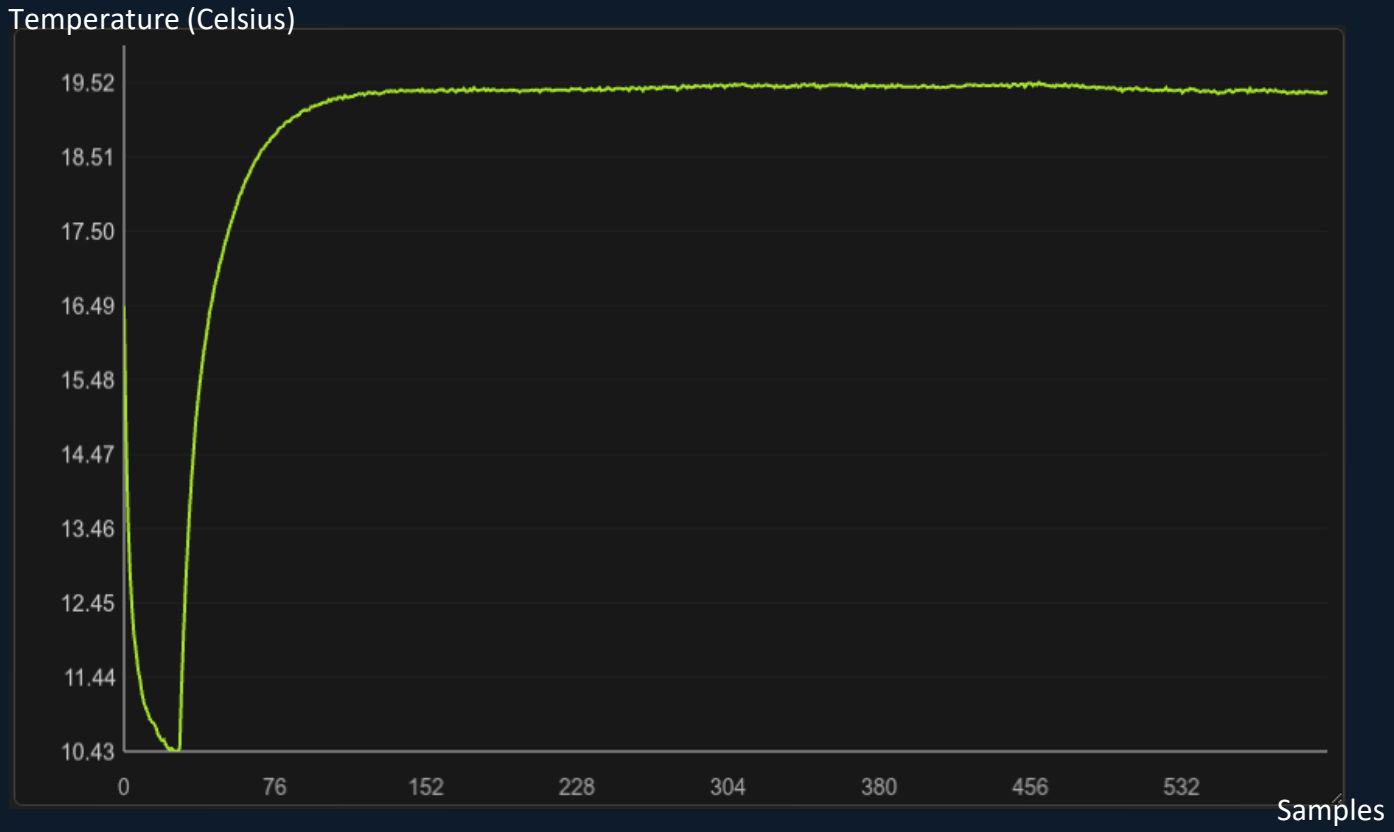

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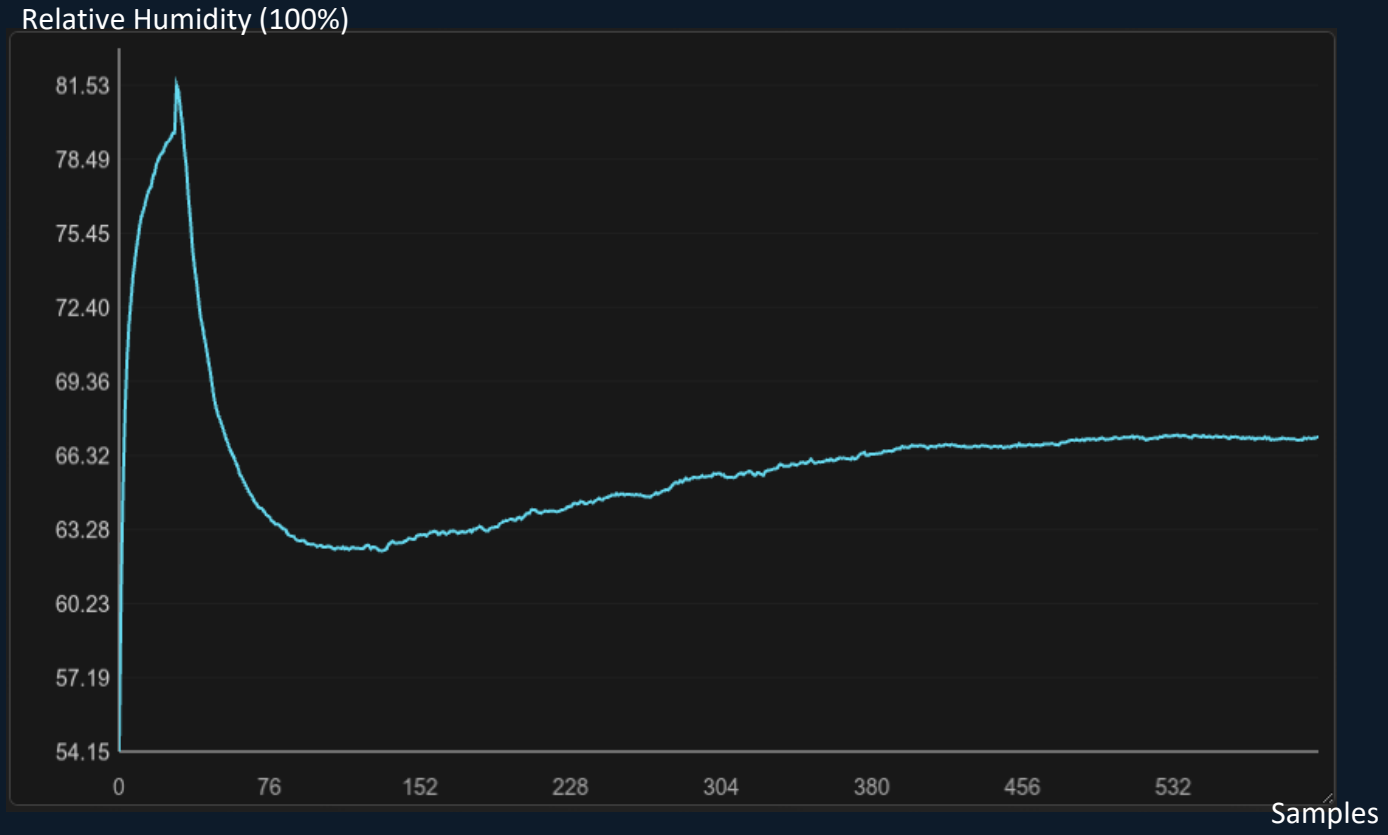


CONSTRAINTS (updated)

Power usage

At least 3 months of autonomous operation with a 3000 mAh LiPo battery.

Every time the system wakes up, it performs the following tasks:

- Wake up the SCD41 and ask for a measurement
- Light Sleep for 5 seconds during the measurement
- Transmit the data with LoRa.
- Go back to Deep Sleep

False alarm prevention

The SCD41's CO2 measurements have strange peaks, filtering of those values is needed, otherwise false pre alarms will happen.

Latency

Even though the observed phenomenon is slow, the need to filter and average measurements to prevent false alarms, increases latency considerably.

When a VERY High CO2 concentration event verifies, the sensor needs to react within 10 minutes (buzzer).

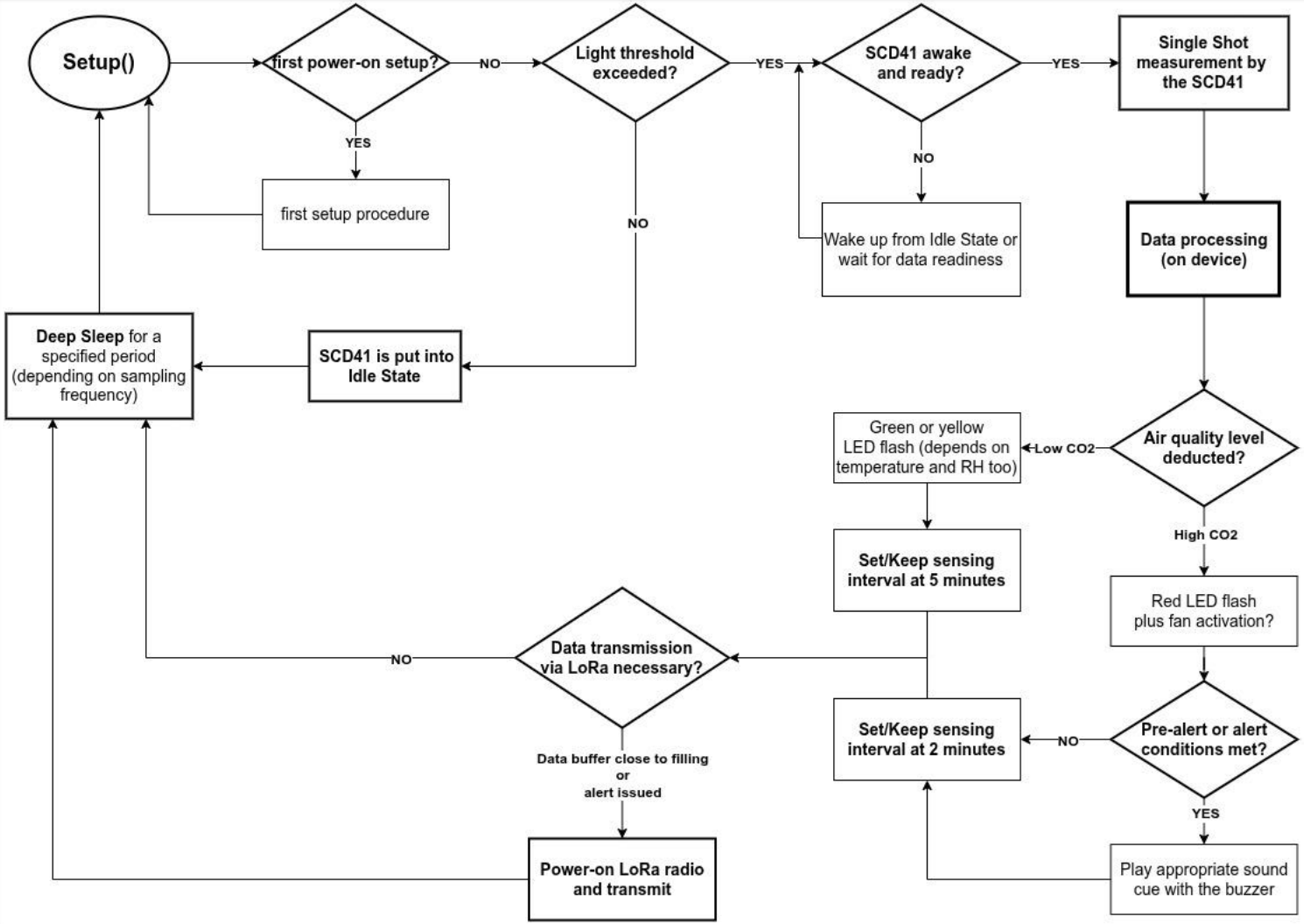
Final Notes

The current prototype operates the SCD41 in two modes:

- **5 minute sampling** interval, at **1,67% duty cycle** (5 seconds of sampling and 295 seconds idling); on the datasheet, it is reported that the sensor requires 0,45 mA on average in this mode.

- **2 minute sampling** interval, at **4,16% duty cycle** (5 seconds of sampling and 115 seconds idling).

Every time a sample is taken, the board has to wait in light sleep for at least 5 seconds (about 1mA required on average).

When the internal buffer of measurements is full, the device transmits all the data at once.

NEXT STEPS (on device)